CSE 198 Lecture #7
Portfolio Optimization
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2 Efficient Frontier

- We Monte-Carlo 15,000 random long-only portfolios of NVDA, AAPL, MSFT, GOOG, AMZN, the gray cloud you see filling in.
- Every portfolio plots a coordinate: (volatility, return)
  - **Closed-form curve (in blue):** The upper boundary of that cloud is the efficient frontier, the best return achievable for each level of risk
- A risk-free asset at r_f = 4.5 % adds a straight line: the Capital Market Line (CML), tangent to the frontier.
- The tangent point is the tangency portfolio, the single portfolio of risky assets with the highest Sharpe ratio (1.40 in this example)
- According to MPT, every rational investor should hold this same tangency portfolio.

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3 Two-Fund Separation

- **Tobin's Two-Fund Separation Theorem**: once a risk-free asset exists, every mean-variance-efficient portfolio is a mix of just two things: the risk-free asset and the same tangency portfolio.
- Formula: efficient portfolio = $\alpha \cdot r\_f + (1 - \alpha) \cdot w\_tangency$. Where α is the cash fraction. Different investors with different risk tolerance pick different α, but they all hold the same stock mix.
- Three example investors:
  - Risk-averse Granny: $\alpha = +0.65$ (65 % in cash, 35 % in the stock mix).
  - Balanced Bob: $\alpha = 0$ (fully invested in the stock mix, no cash).
  - Levered Yousef: $\alpha = -0.5$ (borrows 50 % at r_f and goes 150 % long the stock mix).
- The only difference between Granny, Bob, and Yousef is how much cash they hold or borrow. The risky portion is identical.

4 Non-Stationarity

- MPT assumes return distributions don't change over time ("stationary")
- That assumption fails: NVDA in 2015 is not the same company as NVDA in 2024.
    - 2014-2015: gaming-GPU era. Annualised return ≈ +38 %, volatility ≈ 30 %.
    - 2023-2024: AI-infrastructure era. Annualised return ≈ +113 %, volatility ≈ 49 %.
- Both numbers roughly double. Skewness and kurtosis also shift.
- A single estimate of μ and σ averaged across both windows describes neither regime.

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5 Fat Tails

- MPT (and Sharpe ratio, and standard VaR) treat returns as if they were normally distributed (bell-curve / Gaussian).
- Returns are not Gaussian. They have fat tails: extreme moves happen much more often than the bell curve predicts.
- Concrete demonstration: count NVDA's "4-sigma days" days where the return is more than 4 standard deviations from the mean.
- Under a Gaussian, you'd expect 0.24 such days in 3,772 trading days (roughly one every 125 years).
- NVDA actually had 20 such days ~83× more than the Gaussian predicts.

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6 Black Litterman

- If historical $\hat{\mu}$ is unreliable (Scene E showed why), don't use it as the input to the optimizer.
- Black-Litterman (1992) is the fix: start from a market-implied prior and Bayesian-update it with the investor's explicit views.
  - Prior = the return vector implied by the market portfolio: $\Pi = \lambda \Sigma w_{mkt}$
  - View = an explicit forecast: "NVDA beats the basket by 10 % annually", with a confidence parameter.
  - Posterior μ_{BL} = a weighted blend of prior + view, where the weights depend on the confidence.
  - Sliding the confidence dial from low → high moves the posterior from the prior toward the view.
  - With no confidence → you fall back to the market. With high confidence → you tilt toward what you believe.

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7 Crisis Correlations

- MPT also assumes the correlation matrix is stable. It isn't.
- In a market crisis, all pairwise correlations rush toward +1 (everything moves together)
- Concrete measurement on a 50-stock universe:
  - **Calm regime (2018-2019)**: average pairwise correlation = 0.40.
  - **Crisis regime (Feb-Apr 2020, COVID crash)**: average pairwise correlation = 0.74.

(near-doubling).

- The diversification benefit promised disappears when you need it.
- The portfolio you optimized with calm-regime correlations is materially under-hedged during the crisis.

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7 Problem with Sharpe

- Sharpe ratio penalizes upside and downside equally.
- Two strategies can have identical Sharpe and have wildly different equity curves.
- Example 1-year equity curves.
  - **Strategy A**: occasional big winner day (+20 %).
  - **Strategy B**: occasional big loser day (−20 %).
- Both end up with the **same mean, the same σ, and the same Sharpe = 0.57.**
- Sortino would capture this: A = 1.15 vs B = 0.67.

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8 Spectral Decomposition

- Spectral decomposition: The covariance matrix Σ has a special structure: it can be rewritten as $\Sigma = Q \Lambda Q^T$.
- Q is a rotation matrix whose columns are the eigenvectors of Σ.
- Λ is a diagonal matrix of the eigenvalues of Σ.
- Geometrically: imagine drawing the "1-standard-deviation contour" of a 2D Gaussian cloud. It's an ellipse.
- The eigenvectors point along the ellipse's axes. The eigenvalues are how long those axes are (squared).
- The largest eigenvalue points along the maximum-variance direction (the long axis of the ellipse).
- The smallest eigenvalue points perpendicular to that (the short axis).

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10 Spiked Covariance Model

- Most of the eigenvalues of a sample covariance matrix are noise, they describe randomness.
- **Random-matrix theory predicts the exact shape of the noise floor:** the Marchenko-Pastur (MP) distribution.
- For a 50-stock × 752-day sample ($q = n/T \approx 0.067$), the noise floor sits at $\lambda_+ = (1 + \sqrt{q})^2 \approx 1.58$. Eigenvalues below 1.58 are statistically indistinguishable from random noise.
- A meaningful "signal" axis (like a market factor) shows up as a spike above the noise floor.
  - Example 50-stock returns: 5 eigenvalues sit above the noise floor, dominated by the market factor at $\lambda_1 \approx 17$. The rest of the spectrum is sampling noise.

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10 Ledoit–Wolf Shrinkage

- **Problem:** Sample-covariance eigenvalues are spread out too much: the big ones are too big and the small ones are too small.
- Inverting $\hat{\Sigma}$ reciprocates each eigenvalue ($1/\lambda$). The tiny noisy eigenvalues become huge numbers and explode the portfolio weights.
- The **condition number $\kappa = \lambda_{max} / \lambda_{min}$** measures how unstable the matrix is under inversion.
- For raw sample $\hat{\Sigma}$ of 50-stock returns: $\kappa \approx 288$ (catastrophically unstable).
- **Ledoit-Wolf shrinkage:** blend $\hat{\Sigma}$ with a structured target F. $\Sigma_{LW} = \delta F + (1-\delta) S$. δ is the shrinkage intensity.
- As δ slides from 0 (pure sample) to 1 (pure target), the eigenvalues compress toward each other. **κ drops from 288 to ~23.**

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11 Risk Decomposition

- For any portfolio w, rewrite it in the eigenportfolio basis: $c = Q^T w$.
- Then the portfolio variance breaks into a clean sum: $w^T \Sigma w = \Sigma_i c_i^2 \lambda_i$.
- Each term $c_i^2 \lambda_i$ is the portfolio's exposure to risk axis q_i, weighted by that axis's variance.
- Example: equal-weight portfolio over 50 stocks.
  - Total annualised variance: 0.0331.
  - The first term $c_1^2 \lambda_1$ alone (market-factor risk) contributes ~0.0308 = ~93 % of total.
  - Market-neutralise = force $c_1 = 0$. The dominant variance term vanishes.
  - After market-neutralisation: total variance drops from 0.0331 to 0.0023 **(14× reduction)**.
- We can use this to build market-neutral portfolios: by projecting onto Σ's eigen-basis and explicitly zeroing the market coordinate.

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a. Sortino

- Sortino fixes Sharpe's symmetric-penalty bug flagged in Scene I.
- Replace ordinary volatility σ with downside semi-deviation σ_D: σ_D only measures variance below a target return (typically the risk-free rate or zero).
- Formula: Sortino = $(\mu - r_f) / \sigma_D$.
- $\sigma_D^2 = E[\min(r - r_f, 0)^2]$ *zero on upside days, squared shortfall on downside days*.
- For NVDA (2010-2024): annualised σ = 45.1 %, but annualised σ_D = 30.3 %. Most of NVDA's "volatility" is upside volatility.
- NVDA Sharpe = 0.86. NVDA Sortino = 1.27. Sortino rewards NVDA for having mostly upside vol.

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b. Probabilistic Sharpe Ratio

- A sample Sharpe is a guess at the true Sharpe, with its own statistical uncertainty.
- The Probabilistic Sharpe Ratio (PSR) (Bailey & López de Prado 2012) is a statistical test for "is my true Sharpe really better than some benchmark SR*?"
- PSR = Φ(t-statistic), where Φ is the normal CDF and the t-statistic accounts for skew and kurtosis of the returns.
- The denominator of the t-stat has fat-tail corrections (γ_3 = skew, γ_4 = excess kurtosis) that inflate the standard error when returns are non-Gaussian.
- Example: $n = 48$ monthly returns, sample Sharpe = 1.16, $\gamma_3 = -2.14$, $\gamma_4 = +5.54$.
  - **Under the Gaussian assumption**: PSR = 91.5 % (looks strong).
  - **Under the fat-tail correction**: PSR = 69.4 % (less convincing).

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c. Value-at-Risk & CVaR

- **VaR answers:** "What loss am I unlikely to exceed?"
  - Formally: $VaR_\alpha(r)$ is the loss threshold exceeded only α of the time.
  - "VaR(5 %) = 4.28 %" means "on 95 % of days, my loss is no worse than 4.28 %."
- **CVaR answers:** "How bad is the loss WHEN I exceed VaR?"
  - Formally: $CVaR_\alpha(r) = E[\,-r \mid -r \geq VaR_\alpha\,]$ the average loss conditional on being past VaR.
- For NVDA daily returns (3,772 days):
  - VaR(5 %) = 4.28 %
  - CVaR(5 %) = 6.33 %
- VaR underestimates tail loss by ~48 % here. The average bad day is meaningfully worse than the VaR threshold suggests.

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[Extra] Options Payoff

- For a stock portfolio, P&L is linear in returns: $dPnL = w \cdot dS$.
- For an option, the value $V(S)$ is a non-linear function of the underlying price S. So P&L is non-linear too.
- The first-order approximation is $\Delta \times dS$ (Delta term). The next-order correction is $\frac{1}{2} \Gamma \times (dS)^2$ (Gamma term). Then volatility and time decay terms (Vega and Theta).
- Each Greek is one term in the local approximation of $V(S, \sigma, t)$ in Taylor Expansion
- For a 3-month European call struck at $K = 100$ with $\sigma = 30\,\%$, $r = 4\,\%$, at $S_0 = 105$:
  - $\Delta(S_0) \approx 0.68$ (a \$1 move in S translates to a \$0.68 move in V).
  - $\Gamma(S_0) \approx 0.023$ (Δ itself moves by 0.023 per \$1 move in S).
- **Mean-variance incorrect framework** for portfolio that includes options.

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[Extra] *OptionUniverse* Pipeline

A OptionsUniverse is built in seven stages:

1. Universe construction. calls, puts, spreads, straddles, condors, calendars, earnings-vol trades.
2. Scenario generation. Sample a joint distribution over (S_T, σ_T) at the horizon T. (jump-diffusion)
3. Repricing. For each option × each scenario, compute the P&L. Result: an $N_{scen} \times N_{opt}$ matrix L.
4. Objective. Choose what to optimise: expected P&L, CVaR, Kelly criterion, or any tail metric of L.
5. Constraints. Greek caps ($|\Delta| < 50$, $|\Gamma| < 5$, Vega < 200), margin, liquidity, position limits.
6. Solve. depending on your constraints (stochastic programming)
7. Hedge dynamically. Once live, rebalance Δ and Vega as the underlying and the vol surface move.